

Whats Cookin

User Manual

CS160A Course Project



1. Application Overview

Whats Cookin is a web application that helps users plan meals around the ingredients they already have at home. The application connects a personal digital pantry to a live recipe database, so that recommendations, search results, and recipe details all reflect what a user actually owns.

The Problem It Solves

Home cooks often lose track of what ingredients they have on hand. This leads to duplicate grocery purchases, food that spoils before it is used, and recipes that have to be abandoned partway through because a needed ingredient is missing. Whats Cookin addresses this by keeping a running record of a user's pantry and using it to guide recipe discovery, so users can see in advance which recipes they can already make and which ones need one or two more ingredients.

Intended Users

The application is designed for home cooks who want to reduce food waste and plan meals more efficiently around their existing groceries, including users who are not necessarily experienced in the kitchen.

Main Features

- Account creation and secure login/logout
- A personal pantry with quantities, units, and optional expiry dates
- Automatic expiry status tracking (Fresh, Expiring Soon, Expired)
- Recipe discovery: Recommended Recipes, Everyday Recommended Recipes, and a "Use Soon" section built around ingredients that are close to expiring
- Recipe search and filtering by cuisine and time of day
- A detailed recipe view showing which ingredients are already available and which are missing
- Ingredient substitution suggestions drawn from the user's own pantry

2. Getting Started

Accessing the Application

Whats Cookin is a browser-based application deployed on Vercel. No installation is required. Users can directly open the deployed application URL in a web browser.

Relevant Link(s)

App Demo: <https://whats-cookin-cs160a.vercel.app/>

Github: https://github.com/mrkumli/whats_cookin

Browser and Device Recommendations

The application is built with a responsive layout and works on both desktop and mobile screen sizes. For the best experience, use an up-to-date version of a modern browser such as Chrome, Firefox, Edge, or Safari, with a stable internet connection (required for both account access and recipe data).

Creating an Account

From the Sign Up page, enter an email address, a password, and a confirmation of that password, then submit the form. Passwords must be at least 6 characters. If account creation fails, the form displays a message explaining the likely cause (for example, a password that is too short or an email address that is invalid).

Logging In

From the Login page, enter the account's email and password and submit the form. A successful login redirects the user to the Home page.

Logging Out

The Navigation bar includes a logout control for signed-in users. Selecting it ends the current session and returns the user to a signed-out state.

The image displays two side-by-side form cards. The left card is titled 'Sign up' and contains three input fields: 'Email', 'Password', and 'Confirm password'. Below these fields is a dark green 'Sign up' button. At the bottom of the card, it says 'Already have an account? [Log in](#)'. The right card is titled 'Log in' and contains two input fields: 'Email' and 'Password'. Below these fields is a dark green 'Log in' button. At the bottom of the card, it says 'Don't have an account? [Sign up](#)'.

[Login page and Sign Up page]

3. Pantry Management

The Pantry page is only available to signed-in users. It lists every ingredient currently saved to the user's account and is the foundation for all recipe recommendations.

Opening the Pantry Section

Select "Pantry" from the navigation bar after logging in. The page loads the user's saved pantry items automatically.

Adding Ingredients

At the top of the Pantry page is an add-item form:

- Ingredient name: type into the ingredient field; matching suggestions from a built-in ingredient list appear as autocomplete options.
- Quantity: enter a numeric amount for the ingredient.
- Unit: choose a unit from the dropdown, grouped into Weight (g, kg, oz, lb), Volume (ml, L, tsp, tbsp, cup), and Count (piece, pack, can, bottle, jar, dozen).
- Expiry date: optional; pick a date if the user wants expiry tracking for that item.

Submitting the form validates the entry and, if valid, adds the item to the pantry immediately.

Understanding Expiry Indicators

Each pantry item with an expiry date shows a colored status, calculated automatically from today's date:

- Fresh (green): more than 7 days remain before the expiry date.
- Expiring Soon (orange): 7 or fewer days remain before the expiry date.
- Expired (red): the expiry date has already passed.

Items with no expiry date set show no expiry indicator. A legend at the top of the Pantry page explains each color.

Editing Pantry Items

Each pantry item card can be opened in an edit view, where the name, quantity, unit, and expiry date can be updated. Items can also be removed from the pantry entirely.

How Pantry Information Affects Recipe Recommendations

The pantry is connected live to the recipe features. Any time an ingredient is added, edited, or removed, the Home page's Recommended Recipes and Use Soon sections, as well as the ingredient availability shown on a recipe's details page, update automatically without needing to refresh the page.

Pantry

● Fresh ● Expiring Soon ● Expired



● Apple

6 piece

Expires 1 Sept 2026

● Banana

6 piece

Expires 4 Aug 2026

● Basil

1 jar

Expires 7 Aug 2027

● Bell Pepper

4 piece

Expires 13 Aug 2026

● Black Beans

2 can

Expires 7 Aug 2027

● Black Pepper

1 jar

Expires 6 Aug 2028

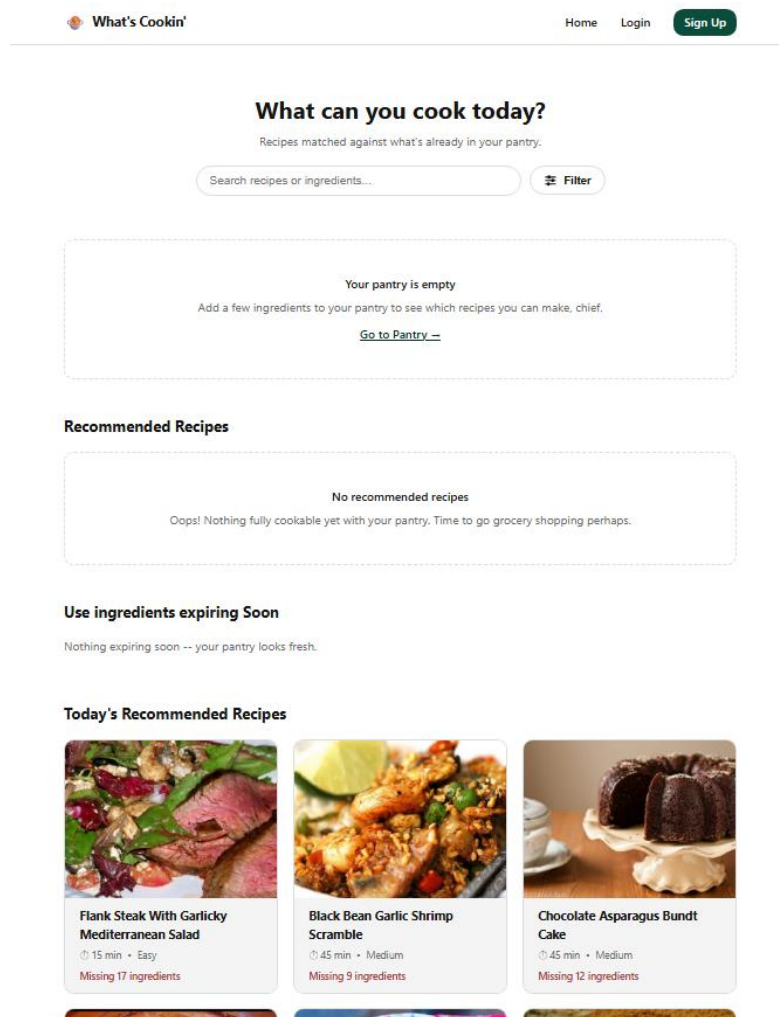
[Pantry page showing item cards with expiry indicators and the add-item form]

4. Recipe Discovery

Homepage

The Home page is the main recipe discovery screen and includes several sections drawn live from the recipe database:

- Recommended Recipes: recipes the signed-in user can already make in full with their current pantry.
- Random Recipes: a broader discovery feed of recipes, shown with a missing-ingredient count on each card even when not fully cookable.
- Use Soon: recipe suggestions built specifically around pantry ingredients that are approaching their expiry date, to help reduce food waste.



[Interactive and Live Homepage]

Recipe Cards

Each recipe card displays the recipe image, title, estimated cook time, difficulty, and (when relevant) the number of missing ingredients or a "Ready to Cook" indicator when every ingredient is already in the pantry.

Recipes Suggested From Pantry Ingredients

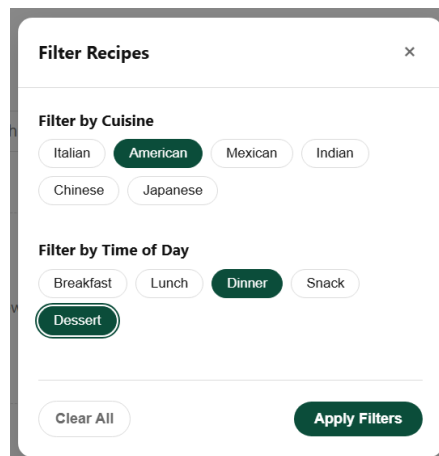
Recommended Recipes and Use Soon are both derived directly from the current pantry: Recommended Recipes narrows a general recipe pool down to only fully cookable recipes, while Use Soon specifically targets ingredients that are expiring soon.

Searching Recipes

Typing into the search bar queries the recipe database directly by title or ingredient. Search results are shown independently of what the pantry can fully make, a recipe will still appear in search results even if ingredients are missing, along with an accurate missing-ingredient count.

Filtering Recipes

A Filter control opens a modal with two filter categories: Cuisine (Italian, American, Mexican, Indian, Chinese, Japanese) and Time of Day (Breakfast, Lunch, Dinner, Snack, Dessert). Multiple options can be selected within a category (matching any of them), and filters across the two categories combine together (a recipe must satisfy both). Active filters appear as removable chips below the search bar and apply across all recipe sections and search results.



[Recipe Filters Pop-Up Window]

5. Recipe Details Page

Opening a Recipe

Selecting any recipe card opens its full details page, showing the recipe image, title, cook time, difficulty, time-of-day category, servings, and cuisine (when available from the recipe source).

Viewing Ingredients

The ingredients list shows every ingredient the recipe requires, each with its quantity.

Available Ingredients

Ingredients the user already has in their pantry are marked with a green indicator. Matching is tolerant of minor wording differences (for example, a pantry item named "Chicken Breast" is recognized as satisfying a recipe ingredient listed as "boneless skinless chicken breasts.")

Missing Ingredients Count

Ingredients not found in the pantry are marked with a red indicator. Recipe cards elsewhere in the application show a running count of these missing ingredients so users can judge at a glance how close they are to being able to cook a given recipe.

Ingredient Substitution Feature

Next to any missing ingredient is a "Substitute" button. Selecting it opens a modal listing possible substitutes for that ingredient, but only the ones the user's pantry actually contains. Choosing a substitute updates the ingredient shown on that recipe's details page (not the original recipe data) and adds a "Substituted for [original ingredient]" note so the change stays visible.

How Users Can Still Cook Recipes With Missing Ingredients

Because missing ingredients are shown individually with their own substitution options, a recipe that is not fully "Ready to Cook" is not hidden or excluded, users can review exactly what is missing, apply pantry-based substitutes where available, and decide for themselves whether to proceed, substitute, or shop for the remainder.

6. Demo Account

A demo account is available for grading and testing purposes so that reviewers do not need to build a pantry from scratch to see the application's full functionality.

Demo Credentials

- **Email:**

- **Password:**

The demo account's pantry is pre-populated with a realistic set of ingredients, deliberately including some that are fresh, some that are expiring soon, some that are already expired, and a few common recipe ingredients left out entirely. Logging in with this account demonstrates:

- Recommended Recipes populated from a real pantry
- Use Soon recommendations driven by expiring ingredients
- Recipes with missing ingredients and accurate missing-ingredient counts
- The ingredient substitution feature with real substitute matches
- Expiry tracking across all three status states (Fresh, Expiring Soon, Expired)

7. Troubleshooting

Issue	Likely Cause / What To Try
Cannot create account	Confirm the password is at least 6 characters and both password fields match. Confirm the email address is correctly formatted and not already registered.
Cannot log in	Confirm the email and password are correct. If the account was created recently, confirm it was created successfully rather than failing silently.
Empty recipe results	This can happen if the recipe search API is temporarily unavailable or the search term is very specific. Try a broader search term or clear active filters, which may be excluding all available results.
API / network problems	The application depends on an external recipe data source and an internet connection. If recipe sections fail to load, check your connection and try again; the application will retry rather than crash.
Pantry not loading	Confirm you are logged in, since the Pantry page requires an active session. If the page stays in a loading state, refresh the browser tab.
API limit reached	Since the application is built as a prototype for an academic resource, the main use remains evaluating the design and workflow of the principles and thus does not own a proper API source to funnel all the requirements. If the API limit is reached and recipes do not appear, kindly wait a few hours for the limit to auto-reset. We apologize for this limitation but it was necessary to sustain the budgeting allocations.